

CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

Some [lectures](#), [job postings](#), and [FAQ](#) are now available. See [update log](#) for update history and [giving](#) for donation. [Contact](#) info is given below.

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PDB Info	STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6	JOB ID: 3757172590
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Title	
PDB ID	8HBV_WITH_8G8W_CARDIOLIPINS
Type	Protein
Experimental Method	Unknown

PDB Manipulation Options:

☒ System pH: [Apply](#)☒ Reading Hetero Chain Residues:

CDL ☒ Rename to [CSML Search](#) **Click this if you want to generate your ligand FF using the PDB coordinate.**

☐ Use CHARMM General Force Field to generate CHARMM top & par files (using [ParamChem](#) service) [?](#)

☐ Use Antechamber to generate CHARMM top & par files

☐ Use OpenFF to generate CHARMM top & par files

☐ Upload CHARMM top & par for hetero chain [?](#)

☒ Protonate/Deprotonate based on selected pH

Ligand reading and its FF generation could be tricky, so please read these common [issues](#).

If this residue exists in the CHARMM topology files ([toppar.str](#)), please rename (or retype) it to the CHARMM residue name. Otherwise, CHARMM will be terminated abnormally.

☒ Terminal group patching: [?](#)

PROA ☐ Cyclic peptide?

☐ Preserve hydrogen coordinates:☐ Mutation:☐ Protonation state:☐ Disulfide bonds:☐ Phosphorylation:☐ Ubiquitylation / SUMOylation:

☐ GPI anchor:☐ Glycosylation / Glycan Ligand(s): ☐ Use CHARMM MC?

"test_distortion.txt" file.

It is faster than the regular run, but carefully check the output

☐ Heme coordination☐ Add Lipid-tail ☐ Peptide Stapling ☐ Add FRET/LRET fluorophore labels ☐ Model LBT-loop(s) ☐ Add MTS reagents: nitroxide spin labels ☐ Add MTS reagents: chemical modifier ☐ Non-standard amino acid / RNA substitution:☐ Lys / Arg PTMs: **Symmetry Operation Options:**

Next Step:
Generate PDB and Orient Molecule

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